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CSE 6015, SMT
Lecture sheet – 8

Language Modeling using N-grams

Assigning a probability to each unit of text. In SMT, a unit of text is a sentence.

Given a sentence e
our task is to compute P(e)

A sentence e containing the word sequence $W_1 W_2 \ldots W_n$

$e \Rightarrow W_1 W_2 \ldots W_n$

$$P(e) = P(W_1 W_2 \ldots W_n)$$

$$= P(W_1)\, P(W_2|W_1)\, P(W_3|W_1 W_2) \ldots P(W_n|W_1 W_2 \ldots W_{n-1})$$

Example:

[illegible]

আমি ভাত খাই।

P(আমি ভাত খাই) = P(আমি) * P(ভাত|আমি) * P(খাই|আমি ভাত)

$P(W_n|W_1 W_2 \ldots W_{n-1})$ calculation:

Data sparsity problem

The N-gram approximation:

N-gram model:

The probability of a word given all previous words is approximated by the probability of the word given the previous N−1 words.

N = 1	unigram model
N = 2	Bigram model
N = 3	Trigram model

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Unigram Model :

Corpus

আমি ভাত খাই-
সে ভাত খায়
তুমি ভাত খাও
আমরা ভাত খাই-

$$P(W_i) = \frac{count(W_i)}{count(W)}$$

Count (W_i) = Total number of W_i words

Count (W) = Total number of any words.

$$P(\text{ভাত}) = \frac{count(\text{ভাত})}{count(\text{all words})} \quad \ldots \text{eq(1)}$$

count (ভাত) = 4

count (all words including ভাত)

= আমি + ভাত + খাই- + সে + খায় + তুমি + খাও + আমরা

= 1 + 4 + 1 + 1 + 1 + 1 + 1 + 1

= 12

From eq (1) $\Rightarrow$

$$P(\text{ভাত}) = \frac{count(\text{ভাত})}{count(\text{all words including ভাত})}$$

$$= \frac{4^{1}}{12_{3}}$$

$$= \frac{1}{3}$$

$$= 0.333$$

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Model :

(3)

$$P(w_n | w_{n-1}) = \frac{count(w_{n-1} w_n)}{\sum_{w} count(w_{n-1} w)}$$

Here, $count(w_{n-1} w)$ denotes the number of occurrences of the sequence $w_{n-1} w_n$.

$count(w_{n-1} w)$ denotes the number of times w_{n-1} occurs before any word w.

So, $count(w_{n-1} w) = count(w_{n-1})$

$$P(w_n | w_{n-1}) = \frac{count(w_{n-1} w_n)}{count(w_{n-1})} \quad \ldots$$

আমি ভাত খাই
তুমি ভাত খাও
তুমি ভাত খাও
আমরা ভাত খাই
আমরা ভাত খাই

$$P(\text{খাই} | \text{ভাত}) = \frac{count(\text{ভাত খাই})}{count(\text{ভাত})} \quad \ldots \text{eq (2)}$$

From [redacted] $count(\text{ভাত খাই}) = 2$

[redacted] $count(\text{ভাত}) = 5$

From eq (2) $\Rightarrow$

$$P(\text{খাই} | \text{ভাত}) = \frac{count(\text{ভাত খাই})}{count(\text{ভাত})} = \frac{2}{5} = 0.4$$

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Trigram Model :

$$P(w_n | w_{n-2} w_{n-1}) = \frac{count(w_{n-2} w_{n-1} w_n)}{\sum_{w} count(w_{n-2} w_{n-1} w)}$$

Since, $\sum_{w} count(w_{n-2} w_{n-1} w) = count(w_{n-2} w_{n-1})$

So, $P(w_n | w_{n-2} w_{n-1}) = \frac{count(w_{n-2} w_{n-1} w_n)}{count(w_{n-2} w_{n-1})}$

corpus

আমাদের স্কুলে যাওয়া উচিত

আমাদের স্কুলে যাওয়া উচিত হয় নাই

আমাদের স্কুলে যাওয়া ঠিক নই।

কখন স্কুলে যাওয়া উচিত।

আমাদের স্কুলে যাওয়ার সিদ্ধান্ত ঠিক ছিল

$$P(\text{উচিত} | \text{স্কুলে যাওয়া}) = \frac{count(\text{স্কুলে যাওয়া উচিত})}{count(\text{স্কুলে যাওয়া})} \quad \ldots\ldots \text{eq}^n (3)$$

count(স্কুলে যাওয়া উচিত) = 3

count(স্কুলে যাওয়া) = 4

From eq^n (3) $\Rightarrow$

$$P(\text{উচিত} | \text{স্কুলে যাওয়া}) = \frac{count(\text{স্কুলে যাওয়া উচিত})}{count(\text{স্কুলে যাওয়া})} = \frac{3}{4}$$

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Bigram	Probability
START all	0.16
all men	0.09
men are	0.24
are equal	0.08

Find P(all men are equal). After that find P(all animals are equal).

Ans:

$$P(\text{all men are equal}) = P(\text{all} \mid \text{START}) * P(\text{men} \mid \text{all}) * P(\text{are} \mid \text{men}) * P(\text{equal} \mid \text{are})$$

$$= 0.16 * 0.09 * 0.24 * 0.08$$

$$= ?$$

$$P(\text{all animals are equal}) = P(\text{all} \mid \text{START}) * P(\text{animals} \mid \text{all}) * P(\text{are} \mid \text{animals}) * P(\text{equal} \mid \text{are})$$

(if zero)

$$= 0$$

If P(animals|all) is not found in corpus, P(all animals are equal) will be zero.

Problem arises

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The task of the language model is not just to give probabilities to those sequences that are present in the corpus, but also to make reasonable estimates when faced with a new sequence.

Smoothing:

The task of distributing some of the probability to unseen or zero-probability sequences is called smoothing.

Algorithm:

Add one to all possible sequences' count
- that occur in corpus.
- those that do not occur.

$$p^*(w_n | w_{n-1}) = \frac{count(w_{n-1} w_n) + 1}{count(w_{n-1}) + V}$$

where V = The number of word types in the vocabulary.

Corpus	
আমি ভাত খাই সে ভাত খায় তুমি ভাত খাও আমরা ভাত খাই তোমরা ভাত খাও	সে ভাত খাইতেছে

$P(\text{সে ভাত খাইতেছে}) = P(\text{সে} | \text{START}) * P(\text{ভাত} | \text{সে}) * P(\text{খাইতেছে} | \text{ভাত})$ ← if zero

$= 0$

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P(আমি | START) = ⟶ count = 1+1

P(ভাত | আমি) = ⟶ count = 1+1

P(খাই | ভাত) = ⟶ count = 2+1

P(চা | START) = ⟶ count = 2+1

P(ভাত | চা) = ⟶ count = 2+1

P(খাব | ভাত) = ⟶ count = 2+1

P(তুমি | START) = ⟶ count = 1+1

P(ভাত | তুমি) = ⟶ count = 1+1

P(খাও | ভাত) = ⟶ count = 1+1

P(আমরা | START) = ⟶ count = 1+1

P(ভাত | আমরা) = ⟶ count = 1+1

P(খাই | ভাত) = ⟶ [illegible]

P(আমরা | START) = ⟶ count = 1+1

P(ভাত | আমরা) = ⟶ count = 1+1

P(খাব | ভাত) =

P(চা | START) =

P(চা | চা) =

P(খাইনা | ভাত) = ⟶ count = 0+1

$$P(\text{খাইনা} \mid \text{ভাত}) = \frac{\text{Count}(\text{ভাত খাইনা}) + 1}{\text{count}(\text{ভাত}) + V}$$

$$= \frac{0+1}{6+14} = \frac{1}{20}$$

$$= 0.05$$

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